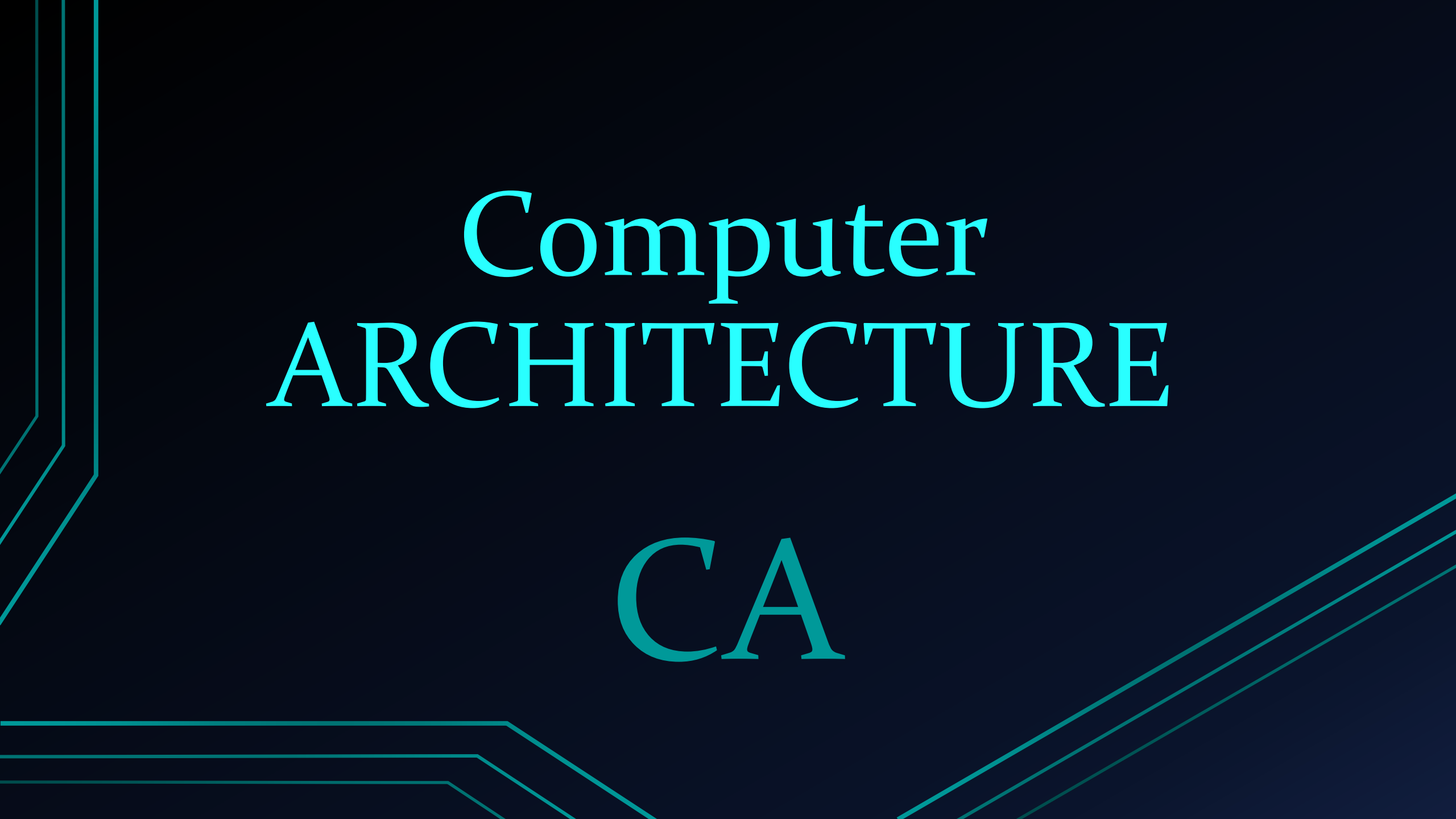




Computer ARCHITECTURE

CA

Introduction

Introduction

- The information processed by a computer can be of different types (text, numbers, etc.)
 - How is information transmitted?
 - How it is represented?
 - How to match the (usual) external representation of information?
 - In what form is this information processed by a computer?

Introduction

- How is information transmitted?
- through what

➤ Data Buses

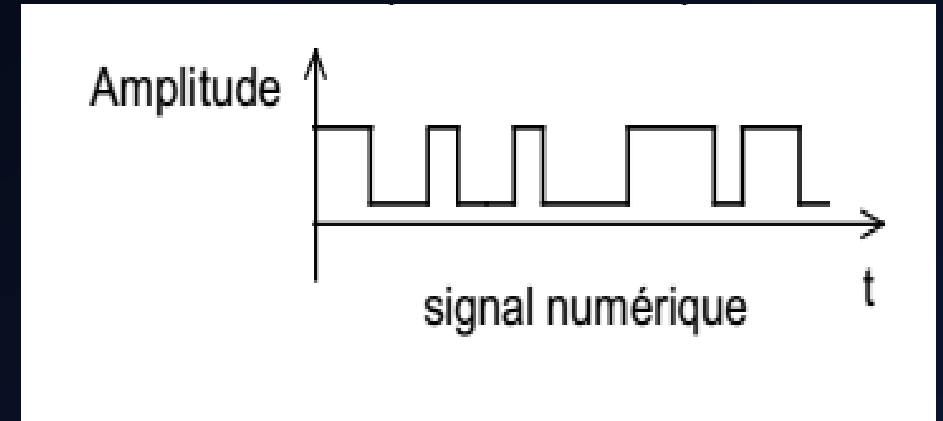
- In what form?

➤ Signal

➤ Definition

- It is the physical medium of information
- An electrical quantity
- Whose modulation represents coded information
- Transported from its source to its destination

➤ **Digital signal** Defined for two time values



Introduction

- In what form is this information processed by a computer?
- How are it represented?
 - By encoding this information using the binary numeral system .

Introduction: basics

- Digital encoding
- Also known as coding of information or digitization, is the process of converting information from its analog form or so-called ***external*** representation (numbers, text, etc.) into another form a digital one or so-called ***internal*** representation of the ***same information*** , following a set of precise rules.
- This conversion involves representing the information using a ***finite*** set of discrete symbols, typically binary digits (0s and 1s).
- Digitization is fundamental in the field of computer science, digital technology, and information processing.

Introduction: basics

- Digital encoding
- Example
- The number 35:
 - 35 is the external representation of the number thirty five
 - The internal representation of 35 will be a sequence of 0 and 1 (100011)

Introduction: basics

- Coding unit
- The components constituting a computer system react, internally, to “all or nothing” signals.
- The two stable states thus defined are represented by the symbols “0” and “1” or by “L” (Low) and “H” (High).
- The information coding unit is an element that can only take the values 0 or 1; the bit.

Introduction: basics

- The bit
- The term bit (lowercase b in notations) means “binary digit”,
- It is the smallest unit of information that can be manipulated by a digital machine.
- A bit can have one of two possible values 0 or 1 in binary numbering.
- These two values correspond to the two fundamental states of a binary system, such as the on/off states of an electronic switch or the presence/absence of an electrical charge.

Introduction: basics

- Binary word
- It typically refers to a sequence of *binary digits* that represents a piece of data or information in a computer.
- In computing, binary words are used to represent various types of data, including numbers, characters, and control information.
- A word is the largest series of bits that a computer can process in a single operation.
- Data is often stored and manipulated in binary words.

Introduction: basics

➤ Binary word

- A sequence of *binary digits* (Each is 0 or 1.)
- The size of these words is most often a multiple of 8.
- Are of fixed length.
 - A unit of information consisting of 16 bits is called a word.
 - A 32-bit unit of information is called a double word (dword).
 - A 64-bit unit of information is called a quad word.
- The size of these words must be distinguished from that of the words processed by the processor (32 bits, 64 bits).
- The size of these two types of words is not necessarily (or theoretically) linked.

Introduction: basics

- Transfer unit
- For data exchanges, elementary information (bits) are manipulated in groups which thus form binary words.
- The transfer unit used for data exchange is called *Byte* or *Octet* .
- The byte or Octet both represent a unit of information composed of 8 bit.
- For example, it allows you to store a character such as a letter or a number.

Introduction

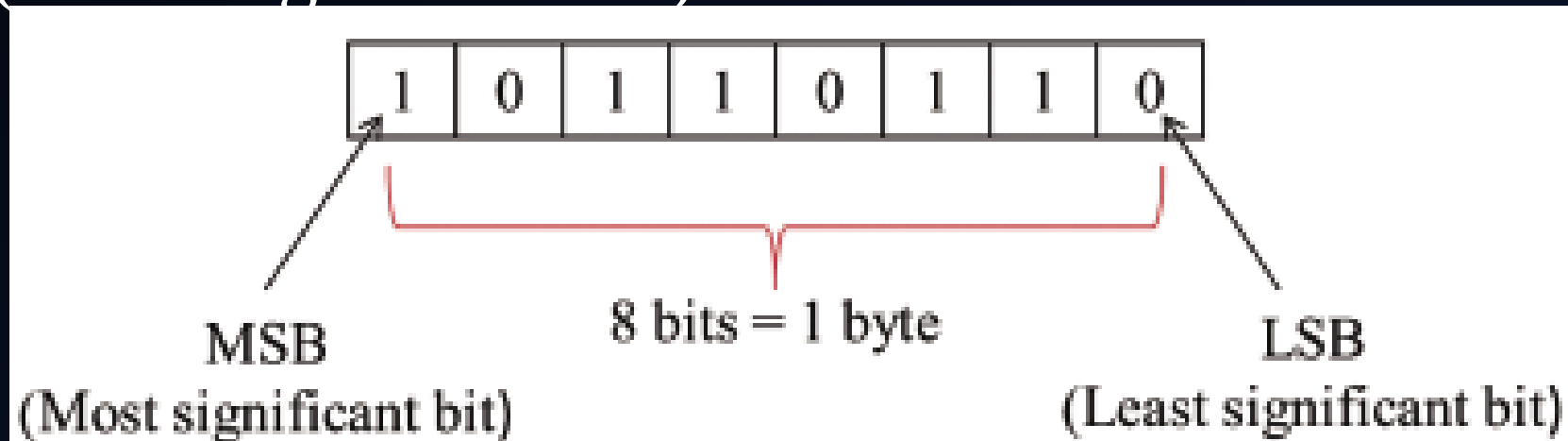
- Multiple units of the byte.
 - Byte = 8 bits
 - “ KiloByte = 2^{10} bytes = 1024 bytes = 1 KB”
 - " MegaByte = 2^{10} KB = 2^{20} bytes = 1024 KB"
 - " GigaByte = 2^{10} MB = 2^{30} bytes = 1024 MB"
 - “ TeraByte = 2^{10} GB = 2^{40} bytes = 1024 GB“
 - “ PetaByte = 2^{10} TB = 2^{50} bytes = 1024 TB“
 - “ExaByte= 2^{10} PB = 2^{60} bytes = 1024 PB”
 - “ ZettaByte = 2^{10} EB = 2^{70} bytes = 1024 EB“
 - “YottaByte= 2^{10} ZB = 2^{80} bytes = 1024 ZB"

Introduction: basic

UNIT	ABBREVIATION	STORAGE
Bit	B	Binary Digit, Single 1 or 0
Nibble	-	4 bits
Byte/Octet	B	8 bits
Kilobyte	KB	1024 bytes
Megabyte	MB	1024 KB
Gigabyte	GB	1024 MB
Terabyte	TB	1024 GB
Petabyte	PB	1024 TB
Exabyte	EB	1024 PB
Zettabyte	ZB	1024 EB
Yottabyte	YB	1024 ZB

Introduction: basics

- To facilitate manipulation, a Byte can be divided into two 4-bit words called nibbles.
- The leftmost bit in a binary number is the most significant bit MSB (Most Significant bit),
- It represents the highest positional value within the number.
- The rightmost bit in a binary number the least significant bit, LSB (Least Significant bit).



Introduction: basic

- Information is measurable with a unit called the bit.
- There is a relationship between the number of bits (I) needed to represent a certain range of values and the total number of *distinct* values (N) that can be represented with those bits.
- The formula $I \text{ (bits)} = \log_2(N)$ where
 - N is the number of possible states gives the quantity of information.
 - $I(\text{bits})$: This represents the number of bits required to represent a given range of values
- The possible states are identified using binary digits.